



Name: _____

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Learning Objectives

- Find critical numbers and classify local extrema
- Determine intervals of increase, decrease, and concavity
- Solve optimization problems using derivatives
- Apply related rates and the Mean Value Theorem

For optimization: define variables, write the objective function, apply constraints, then differentiate and solve.

1. Find all critical numbers of $f(x)$. Then classify each as a local max, min, or neither.

$$f(x) = x^3 - 3x^2 - 9x + 5$$

Answer: _____

2. Find the intervals where $f(x)$ is increasing and decreasing.

$$f(x) = 2x^3 - 12x + 1$$

Answer: _____

3. Find the absolute maximum and minimum of $f(x)$ on the closed interval $[0, 4]$.

$$f(x) = x^3 - 6x^2 + 8 \quad \text{on } [0, 4]$$

Answer: _____

4. Determine the concavity of $f(x)$ and find all inflection points.

$$f(x) = x^4 - 4x^3$$

Answer: _____

5. Optimization: A farmer has 200 m of fencing to enclose a rectangular plot. Find the dimensions that maximize the area.

$$A = x(100 - x), \quad 0 \leq x \leq 100$$

Answer: _____

6. Related Rates: A ladder 10 m long leans against a wall. The bottom slides away at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?

$$x^2 + y^2 = 100, \quad \frac{dx}{dt} = 2$$

Answer: _____



7. Use the Mean Value Theorem. Find c in $(1, 3)$ satisfying the MVT conclusion.

$$f(x) = x^2 - 2x \quad \text{on } [1, 3]$$

Answer: _____

8. Apply L'Hôpital's Rule to evaluate the limit.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

Answer: _____

9. Find the linear approximation (tangent line approximation) of $f(x)$ at $x = 0$.

$$f(x) = \sqrt{1+x} \quad \text{at } a = 0$$

Answer: _____

10. Optimization: A box with a square base and open top must hold 32 cm^3 . Find dimensions minimizing surface area.

$$S = x^2 + \frac{4 \cdot 32}{x}, \quad V = x^2 h = 32$$

Answer: _____





Problems 5 and 10 are optimization — classic exam problems. Problem 6 (related rates): emphasize implicit differentiation with respect to t . Problem 8 (L'Hôpital's): verify 0/0 indeterminate form first.

Solutions

1. Find all critical numbers of $f(x)$. Then classify each as a local max, min, or neither.

$$f(x) = x^3 - 3x^2 - 9x + 5$$

$$\rightarrow f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1).$$

$$\rightarrow \text{Critical numbers: } x = -1 \text{ and } x = 3.$$

$$\rightarrow f''(x) = 6x - 6.$$

$$\rightarrow f''(-1) = -12 < 0 \rightarrow \text{local max. } f''(3) = 12 > 0 \rightarrow \text{local min.}$$

Answer: $x = -1$ (local max), $x = 3$ (local min)

2. Find the intervals where $f(x)$ is increasing and decreasing.

$$f(x) = 2x^3 - 12x + 1$$

$$\rightarrow f'(x) = 6x^2 - 12 = 6(x^2 - 2).$$

$$\rightarrow f'(x) = 0: x = \pm\sqrt{2}.$$

$$\rightarrow f'(x) > 0 \text{ (increasing) when } |x| > \sqrt{2}.$$

$$\rightarrow f'(x) < 0 \text{ (decreasing) on } (-\sqrt{2}, \sqrt{2}).$$

Answer: Increasing on $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

3. Find the absolute maximum and minimum of $f(x)$ on the closed interval $[0, 4]$.

$$f(x) = x^3 - 6x^2 + 8 \text{ on } [0, 4]$$

$$\rightarrow f'(x) = 3x^2 - 12x = 3x(x-4).$$

$$\rightarrow \text{Critical points in } [0, 4]: x = 0 \text{ and } x = 4 \text{ (endpoints).}$$

$$\rightarrow \text{Also } x = 0 \text{ is a critical number } (f'(0)=0) \text{ — it's an endpoint.}$$

$$\rightarrow f(0) = 8, f(4) = 64 - 96 + 8 = -24.$$

$$\rightarrow \text{Also check } x = 4 \text{ is endpoint. Abs max} = 8, \text{ abs min} = -24.$$

Answer: Abs max: $f(0) = 8$; Abs min: $f(4) = -24$

4. Determine the concavity of $f(x)$ and find all inflection points.

$$f(x) = x^4 - 4x^3$$

$$\rightarrow f'(x) = 4x^3 - 12x^2.$$

$$\rightarrow f''(x) = 12x^2 - 24x = 12x(x-2).$$

$$\rightarrow f''(x) = 0 \text{ at } x = 0 \text{ and } x = 2.$$

$$\rightarrow f''(x) \text{ changes sign at both — inflection points at } x = 0 \text{ and } x = 2.$$

Answer: Inflection at $x = 0$ and $x = 2$



5. Optimization: A farmer has 200 m of fencing to enclose a rectangular plot. Find the dimensions that maximize the area.

$$A = x(100 - x), \quad 0 \leq x \leq 100$$

→ Perimeter constraint: $2x + 2y = 200$, so $y = 100 - x$.

→ Area: $A(x) = x(100 - x) = 100x - x^2$.

→ $A'(x) = 100 - 2x = 0 \rightarrow x = 50$.

→ Both sides equal 50 m. Max area = $50 \cdot 50 = 2500 \text{ m}^2$.

Answer: $x = 50 \text{ m}$, $y = 50 \text{ m}$, $A_{\max} = 2500 \text{ m}^2$

6. Related Rates: A ladder 10 m long leans against a wall. The bottom slides away at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?

$$x^2 + y^2 = 100, \quad \frac{dx}{dt} = 2$$

→ Differentiate $x^2 + y^2 = 100$ with respect to t .

→ $2x(dx/dt) + 2y(dy/dt) = 0$.

→ When $x = 6$: $y = \sqrt{100 - 36} = 8$.

→ $2(6)(2) + 2(8)(dy/dt) = 0 \rightarrow dy/dt = -24/16 = -3/2 \text{ m/s}$.

Answer: $\frac{dy}{dt} = -\frac{3}{2} \text{ m/s}$

7. Use the Mean Value Theorem. Find c in $(1, 3)$ satisfying the MVT conclusion.

$$f(x) = x^2 - 2x \quad \text{on } [1, 3]$$

→ MVT: $f'(c) = [f(3) - f(1)] / (3 - 1)$.

→ $f(3) = 9 - 6 = 3$. $f(1) = 1 - 2 = -1$.

→ Average rate: $(3 - (-1)) / 2 = 2$.

→ $f'(x) = 2x - 2 = 2 \rightarrow x = 2$. So $c = 2 \in (1, 3)$. ✓

Answer: $c = 2$

8. Apply L'Hôpital's Rule to evaluate the limit.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

→ Form is $0/0$ at $x = 0$, so L'Hôpital's Rule applies.

→ Differentiate numerator: $d/dx[e^x - 1] = e^x$.

→ Differentiate denominator: $d/dx[x] = 1$.

→ Limit = $\lim(e^x/1) = e^0 = 1$.

Answer: $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

9. Find the linear approximation (tangent line approximation) of $f(x)$ at $x = 0$.

$$f(x) = \sqrt{1+x} \quad \text{at } a = 0$$

→ $L(x) = f(a) + f'(a)(x - a)$.

→ $f(0) = 1$. $f'(x) = 1/(2\sqrt{1+x}) \rightarrow f'(0) = 1/2$.

→ $L(x) = 1 + (1/2)(x - 0) = 1 + x/2$.

Answer: $L(x) = 1 + \frac{x}{2}$



10. Optimization: A box with a square base and open top must hold 32 cm^3 . Find dimensions minimizing surface area.

$$S = x^2 + \frac{4 \cdot 32}{x}, \quad V = x^2 h = 32$$

→ Volume: $x^2 h = 32$, so $h = 32/x^2$.

→ Surface area (open top): $S = x^2 + 4xh = x^2 + 128/x$.

→ $S'(x) = 2x - 128/x^2 = 0 \rightarrow 2x^3 = 128 \rightarrow x = 4$.

→ $h = 32/16 = 2$. Box: $4 \text{ cm} \times 4 \text{ cm}$ base, 2 cm tall.

Answer: $x = 4 \text{ cm}$, $h = 2 \text{ cm}$

